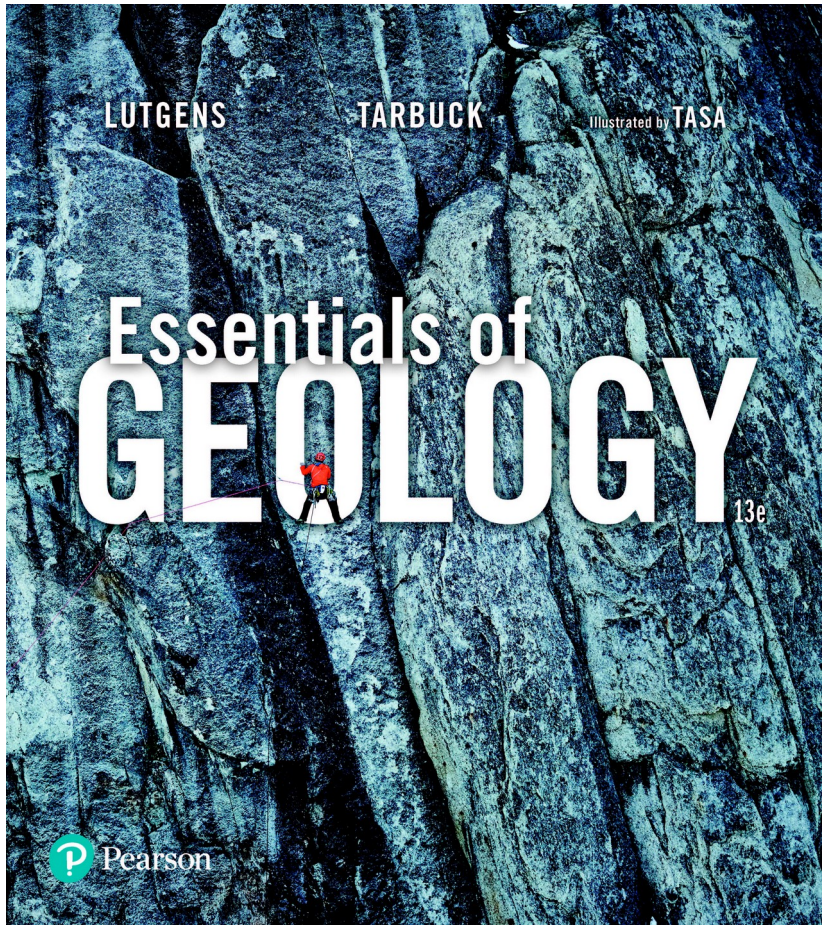
Lecture 3 Outline

Essentials of Geology

Thirteenth Edition



Matter & Minerals

Merry Wilson
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Learning Objectives

- **LO 3.1** List the main characteristics that an Earth material must possess to be considered a mineral and describe each characteristic.
- **LO 3.2** Compare and contrast the three primary particles contained in atoms.
- **LO 3.3** Distinguish among ionic bonds, covalent bonds, and metallic bonds.
- **LO 3.4** List and describe the properties used in mineral identification.
- **LO 3.5** Explain how minerals are classified and name the most abundant mineral group in Earth's crust.

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Learning Objectives (cont.)

- **LO 3.6** Sketch the silicon–oxygen tetrahedron and explain how this fundamental building block joins together to form various silicate structures.
- **LO 3.7** Compare and contrast the light (nonferromagnesian) silicates with the dark (ferromagnesian) silicates and list three common minerals in each group.
- **LO 3.8** List the common nonsilicate minerals and explain why each is important.
- **LO 3.9** Discuss Earth's mineral resources in terms of renewability. Differentiate between mineral resources and ore deposits.

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3.1 Minerals: Building Blocks of Rocks

- Definition of a mineral
  - Naturally occurring
  - Inorganic
  - Solid
  - Ordered internal molecular structure
  - Definite chemical composition
- Definition of a rock
  - A solid aggregate or mass of minerals

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3.2 Atoms: Building Blocks of Minerals

- Atomic structure
  - Central region called the nucleus
    - Consists of protons (positive charges) and neutrons (neutral charges)
  - Electrons
    - Negatively charged particles that surround the nucleus
    - Located in discrete energy levels called shells

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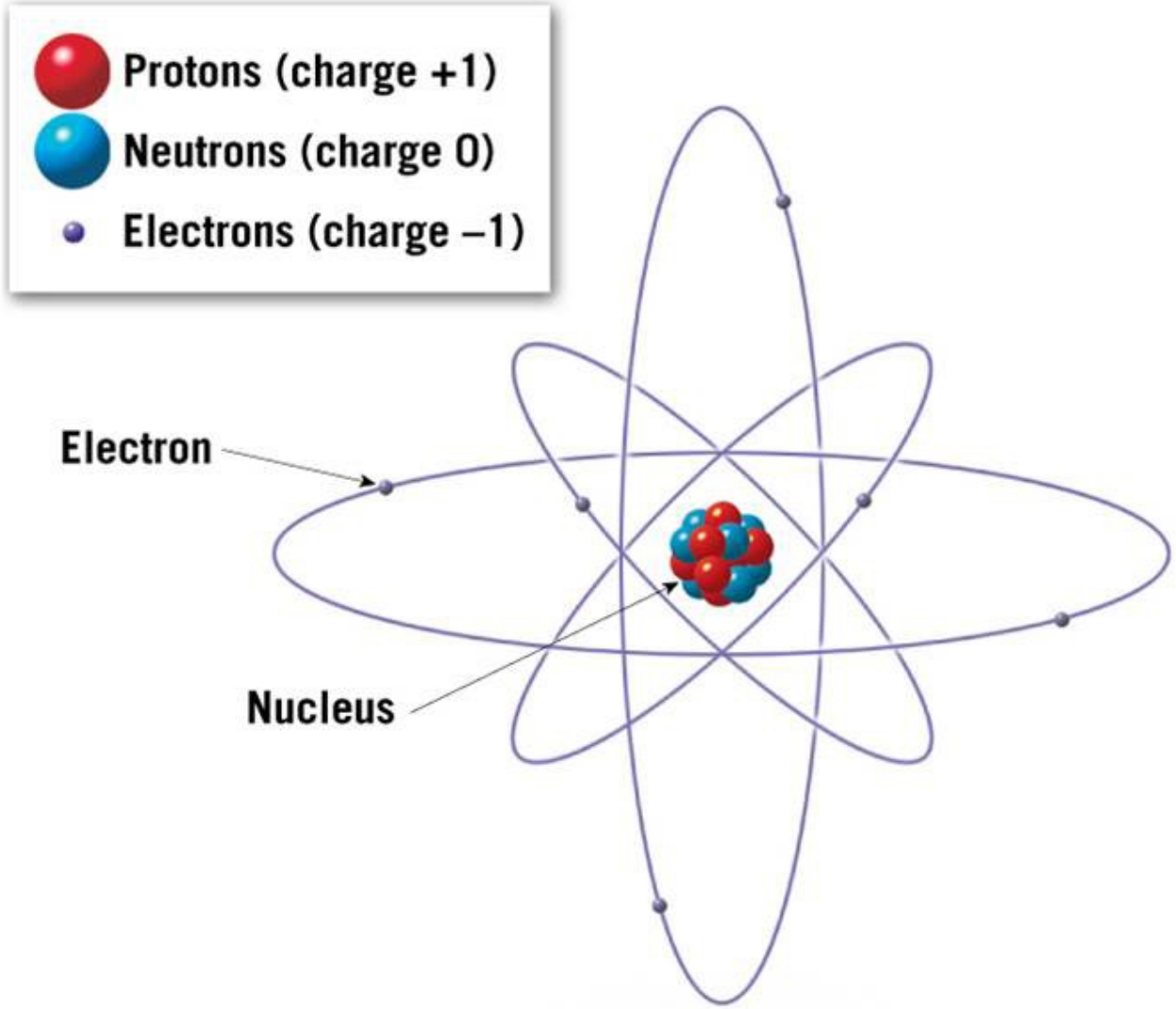
Idealized Structure of an Atom



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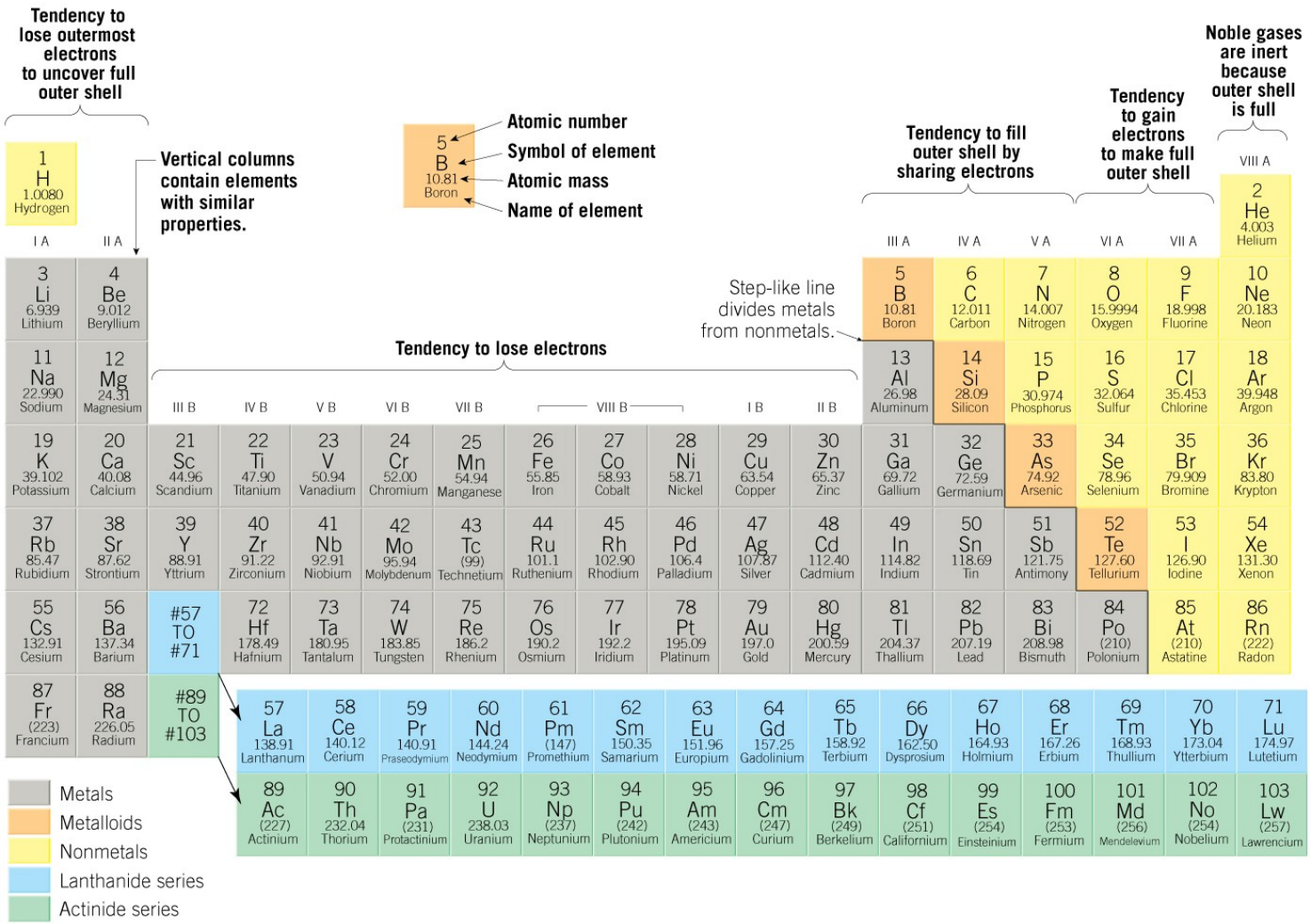
Elements: Defined by Their Number of Protons



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3.3 Why Atoms Bond

- Chemical bonding
  - Formation of a compound by combining two or more elements by transferring or sharing electrons
- Ionic bonding
  - Atoms gain or lose outermost (valence) electrons to form ions
  - Ionic compounds consist of an orderly arrangement of oppositely charged ions

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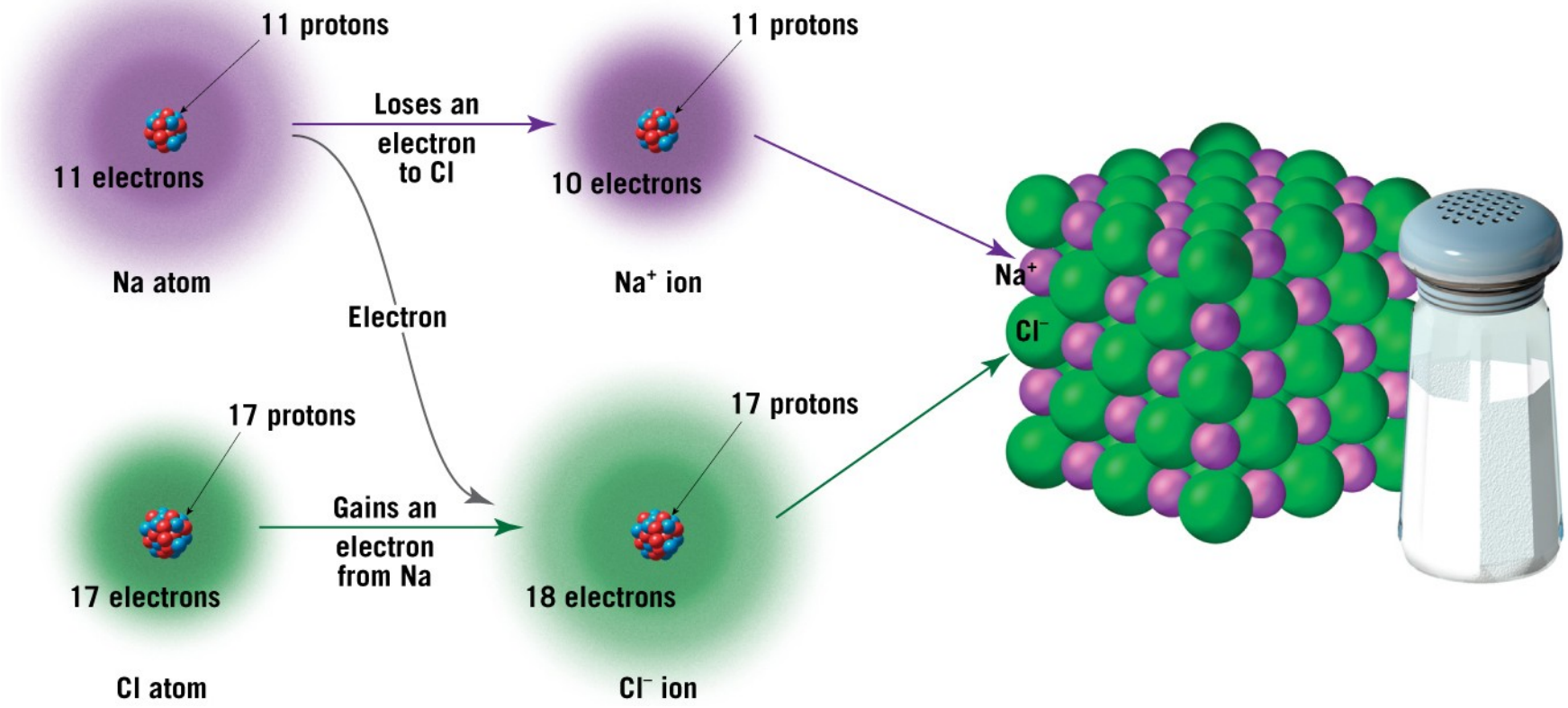
Halite (NaCl): An Example of Ionic Bonding

A. The transfer of an electron from a sodium (Na) atom to a chlorine (Cl) atom leads to the formation of a Na^+ ion and a Cl^- ion.

B. The arrangement of Na^+ and Cl^- in the solid ionic compound sodium chloride (NaCl), table salt.



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3.3 Why Atoms Bond

- Covalent bonding
  - Atoms share electrons to achieve electrical neutrality.
  - Covalent compounds are generally stronger than ionic bonds.
  - Both ionic and covalent bonds typically occur in the same compound (bonds are seldom 100 percent ionic or covalent in character).

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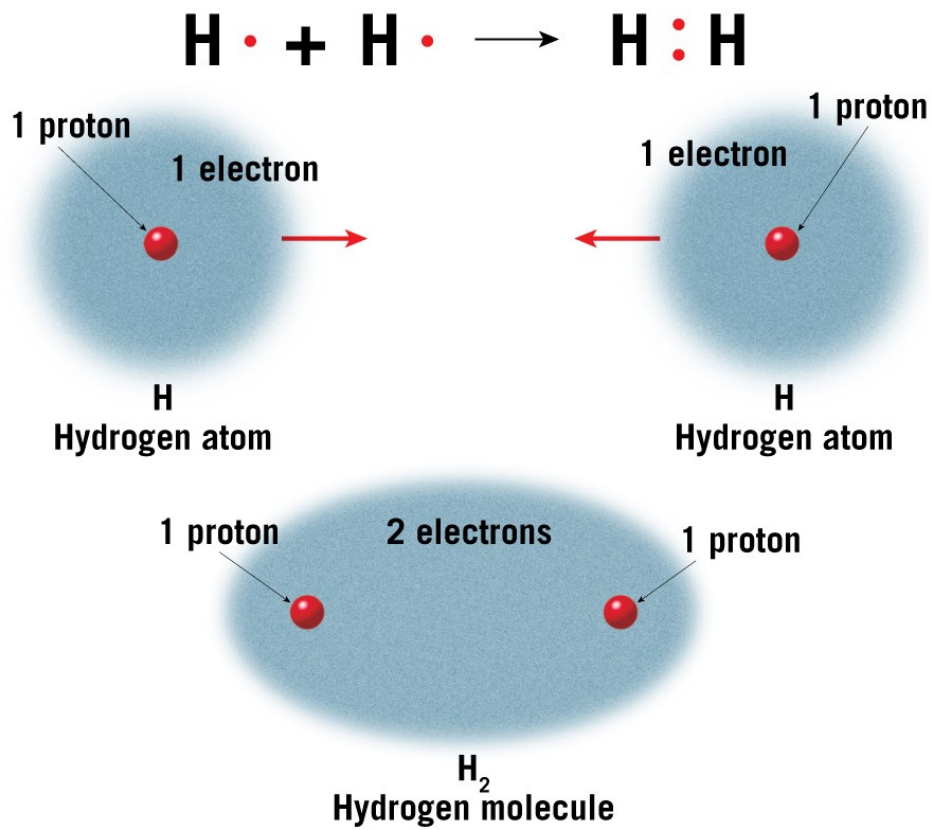
Covalent Bonding: Sharing of Valence Electrons



Two hydrogen atoms combine to form a hydrogen molecule, held together by the attraction of oppositely charged particles—positively charged protons in each nucleus and negatively charged electrons that surround these nuclei.

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Composition of Minerals

- Metallic bonding
  - Valence electrons are free to migrate among atoms
  - Weaker and less common than ionic or covalent bonds

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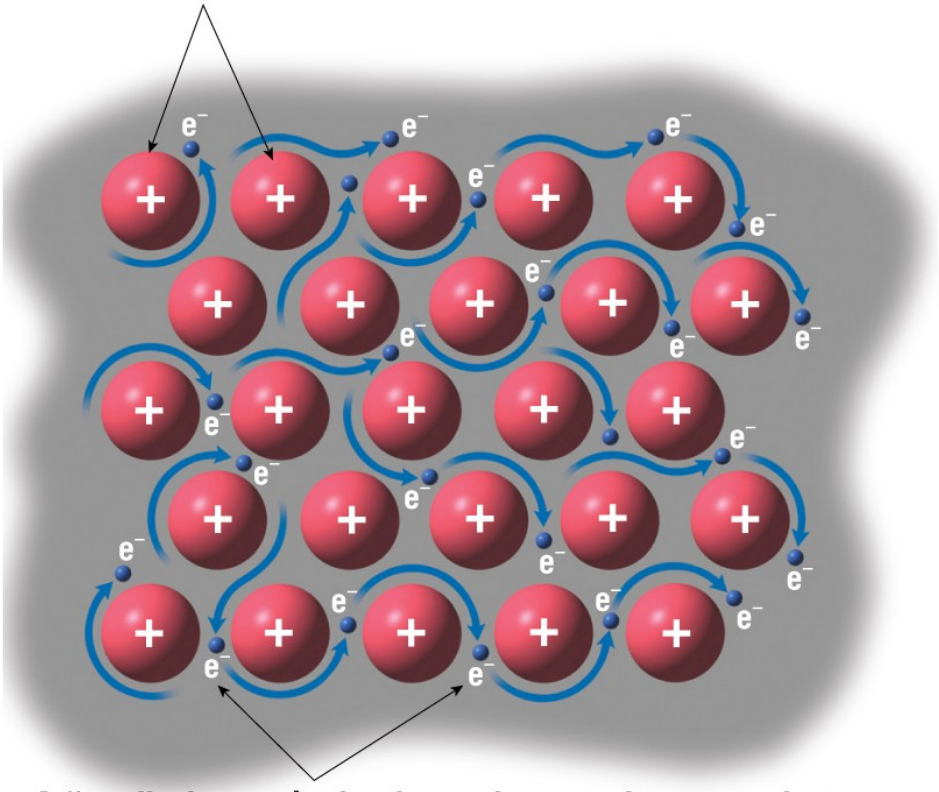
Metallic Bonding

The central core of each metallic atom, which has an overall positive charge, consists of the nucleus and inner electrons.



A "sea" of negatively charged outer electrons, that are free to move throughout the structure, surrounds the metallic atoms.

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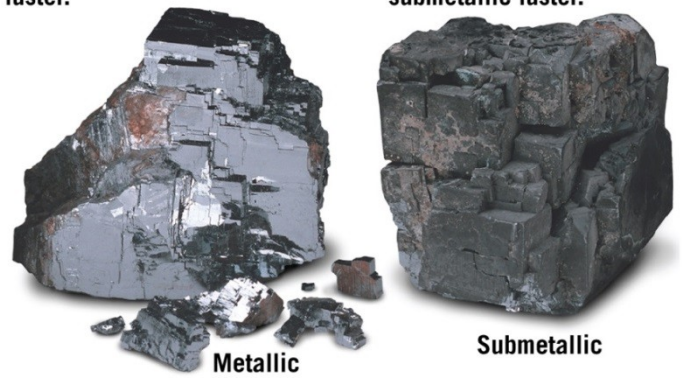
3.4 Properties of Minerals

- Optical Properties
  - Luster
    - Appearance of a mineral in reflected light
    - Two basic categories
      - Metallic
      - Nonmetallic
    - Other terms are used to further describe luster such as vitreous, silky, or earthy.



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3.4 Properties of Minerals (cont.)

- Optical Properties
  - Color
    - Generally an unreliable diagnostic property to use for mineral identification
    - Often highly variable for a given mineral because of slight changes in mineral chemistry
    - Exotic colorations of some minerals produce gemstones

Tutorial

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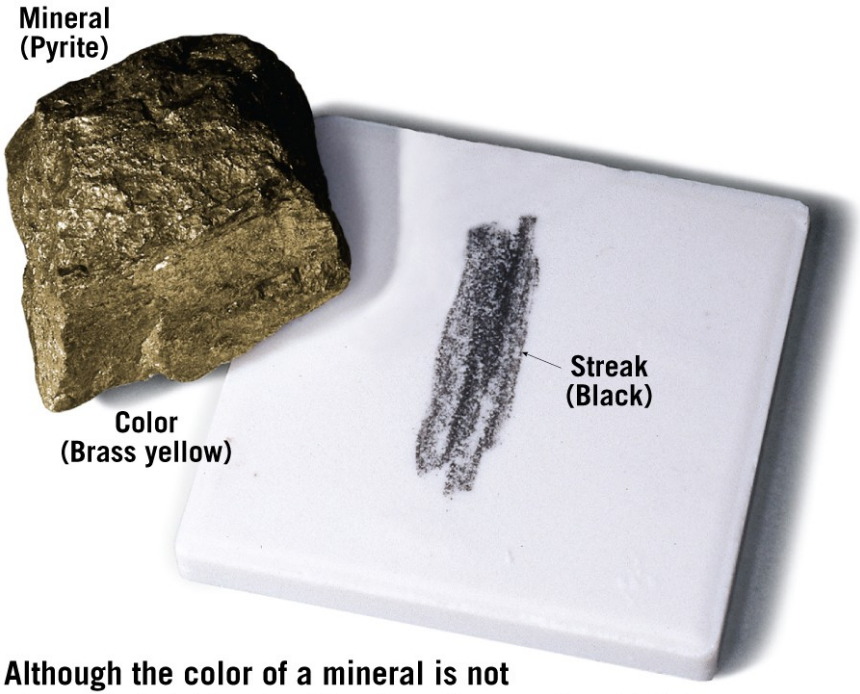
3.4 Properties of Minerals (cont.)

- Optical Properties
  - Streak
    - Color of a mineral in its powdered form
    - Helpful in distinguishing different forms of the same mineral

Video



Although the color of a mineral is not always helpful in identification, the streak, which is the color of the powdered mineral, can be very useful.

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3.4 Properties of Minerals (cont.)

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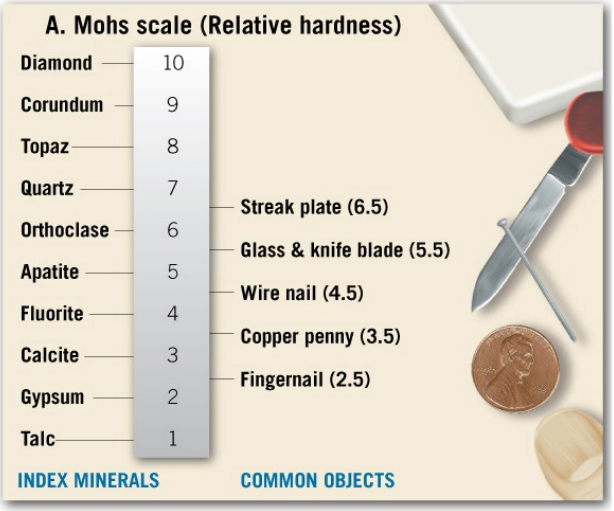
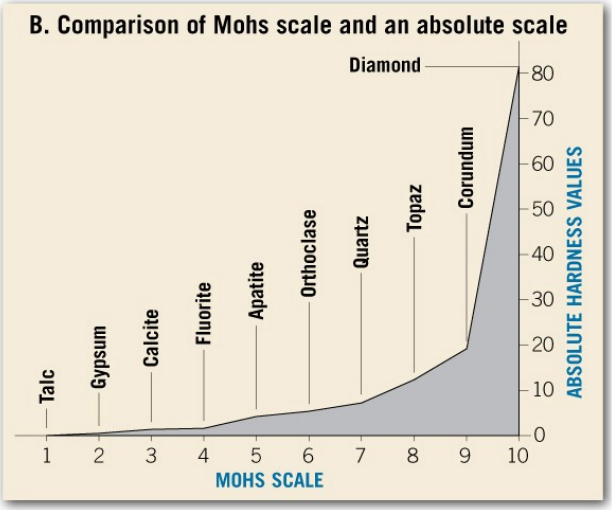
- Mineral Strength
  - Hardness
    - Resistance of a mineral to abrasion or scratching
    - All minerals are compared to a standard scale called the Mohs scale of hardness





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3.4 Properties of Minerals (cont.)

- Mineral Strength
  - Cleavage
    - Tendency to break along planes of weak bonding
    - Produces flat, shiny surfaces
    - Described by resulting geometric shapes
      - Number of planes
      - Angles between adjacent planes

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Common Cleavage Directions

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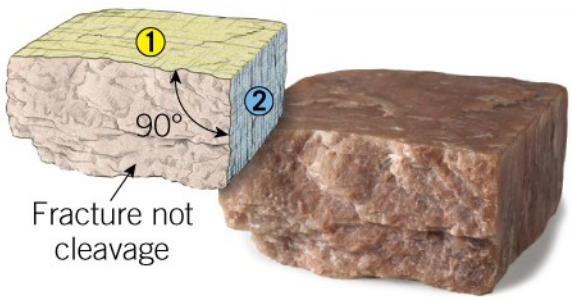
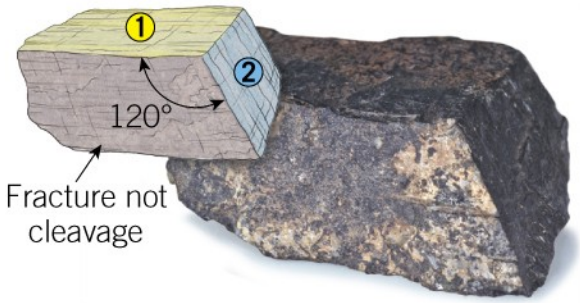
A. Cleavage in one direction. Example: Muscovite

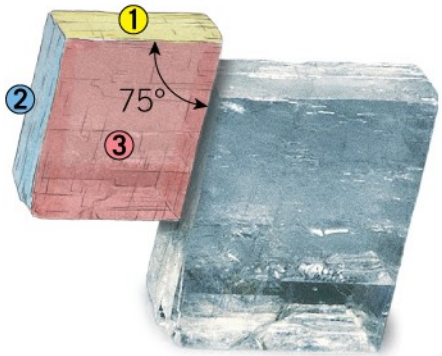


B. Cleavage in two directions at 90° angles. Example: Feldspar

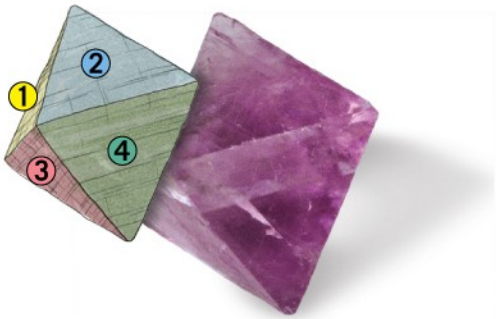


C. Cleavage in two directions not at 90° angles. Example: Hornblende

D. Cleavage in three directions at 90° angles. Example: Halite



E. Cleavage in three directions not at 90° angles. Example: Calcite



F. Cleavage in four directions. Example: Fluorite

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3.4 Properties of Minerals

- Specific gravity
  - Ratio of the weight of a mineral to the weight of an equal volume of water
  - Average value is approximately 2.7

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3.4 Properties of Minerals (cont.)

- Other properties
  - Magnetism
  - Reaction to hydrochloric acid
  - Malleability
  - Double refraction
  - Taste
  - Smell
  - Elasticity

Video

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3.5 Mineral Groups

- Nearly 4000 minerals have been identified on Earth
- Rock-forming minerals
  - Common minerals that make up most of the rocks of Earth's crust
  - Only a few dozen members
  - Composed mainly of the eight elements that make up more than 98 percent of the continental crust

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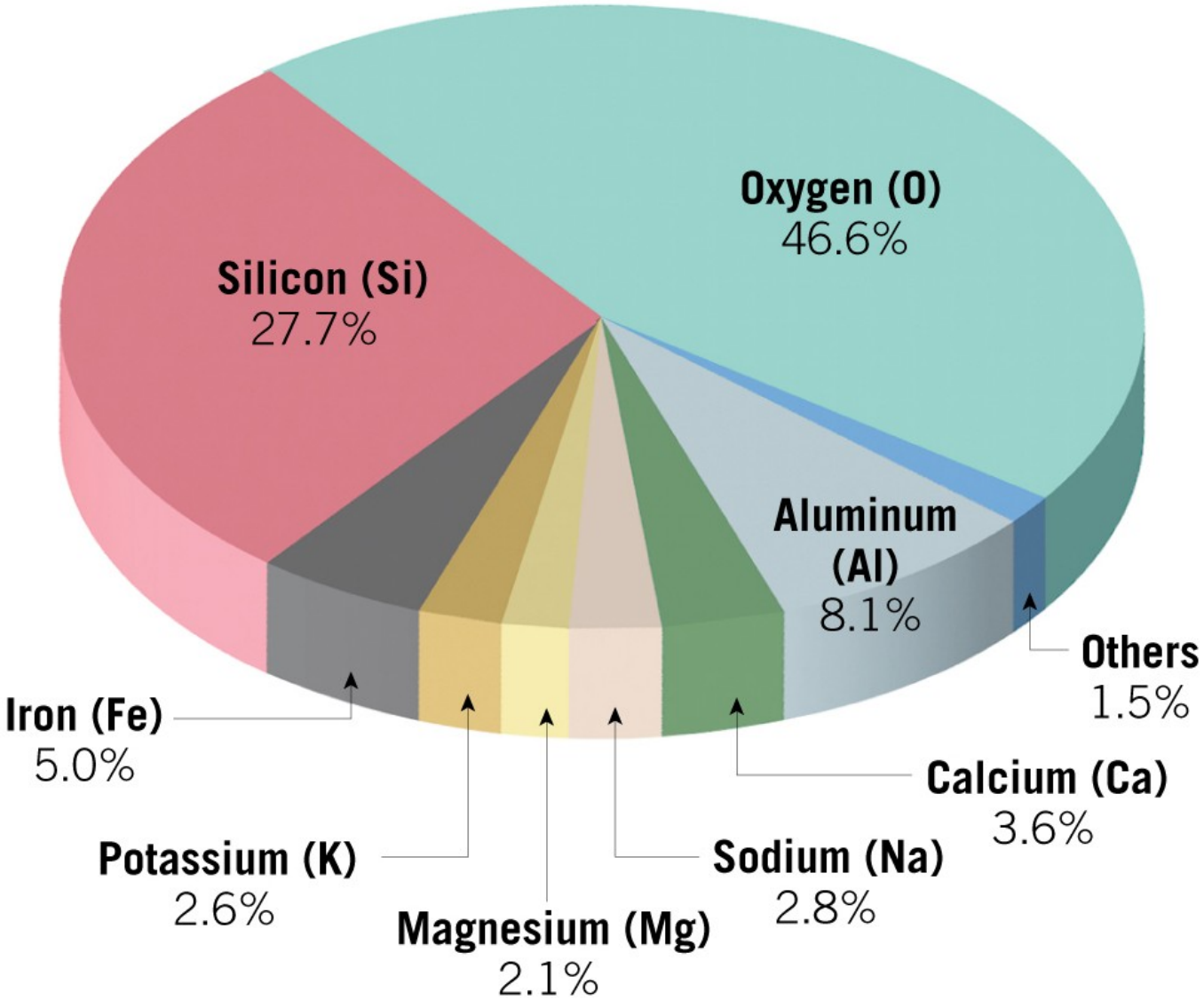
Elemental Abundances in Continental Crust



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3.6 The Silicates

- Silicates
  - Most important mineral group
    - Comprise most of the rock-forming minerals
    - Very abundant as a result of large amounts of silicon and oxygen in Earth's crust
  - Basic building block is the silicon–oxygen tetrahedron molecule
    - Four oxygen ions surrounding a much smaller silicon ion

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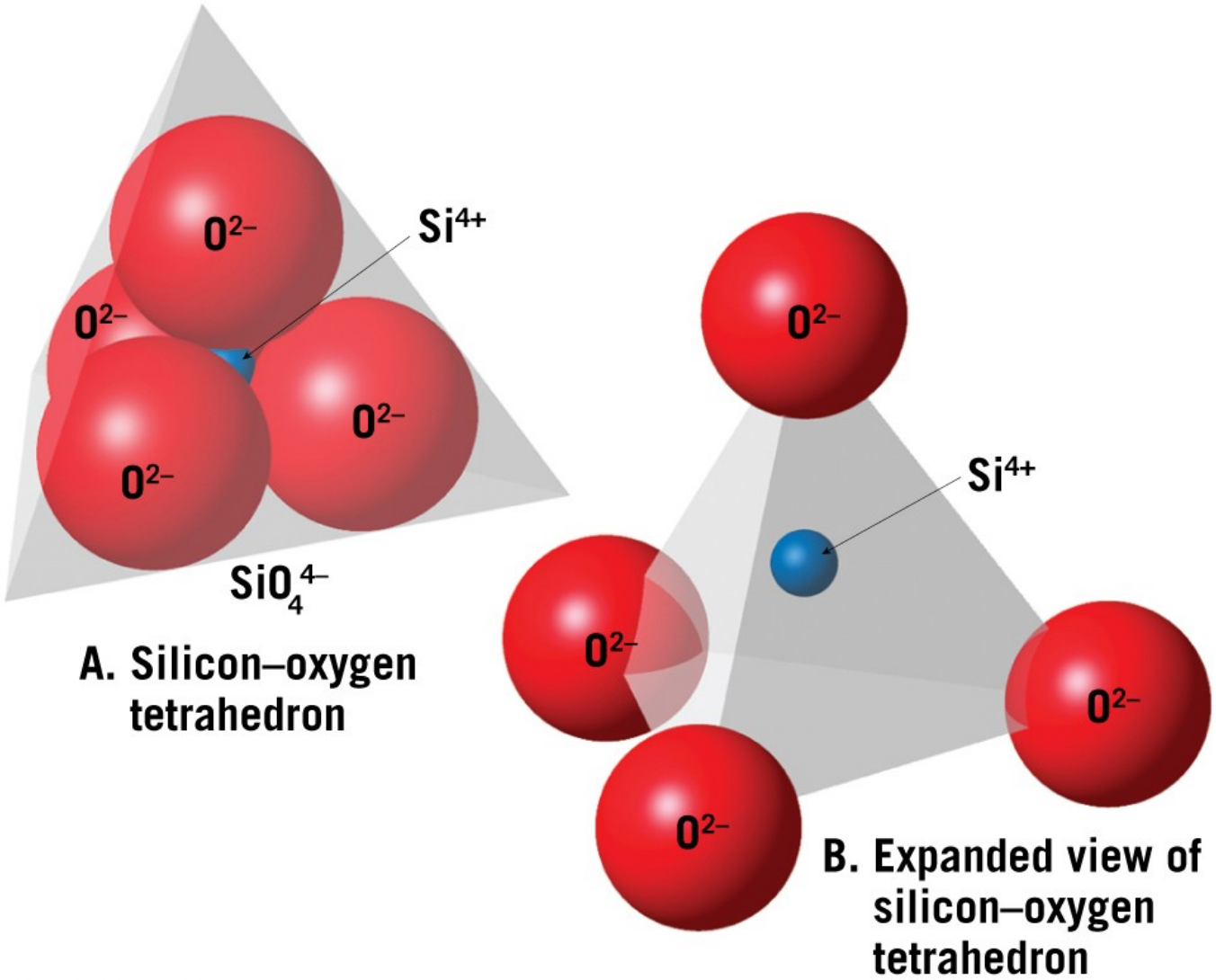
Two Illustrations of the Si–O Tetrahedron



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3.6 The Silicates

- Silicate structures
  - Single tetrahedra are linked together to form various structures including
    - Isolated tetrahedra
    - Ring structures
    - Single- and double-chain structures
    - Sheet or layered structures
    - Complex three-dimensional structures

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Five Basic Silicate Structures

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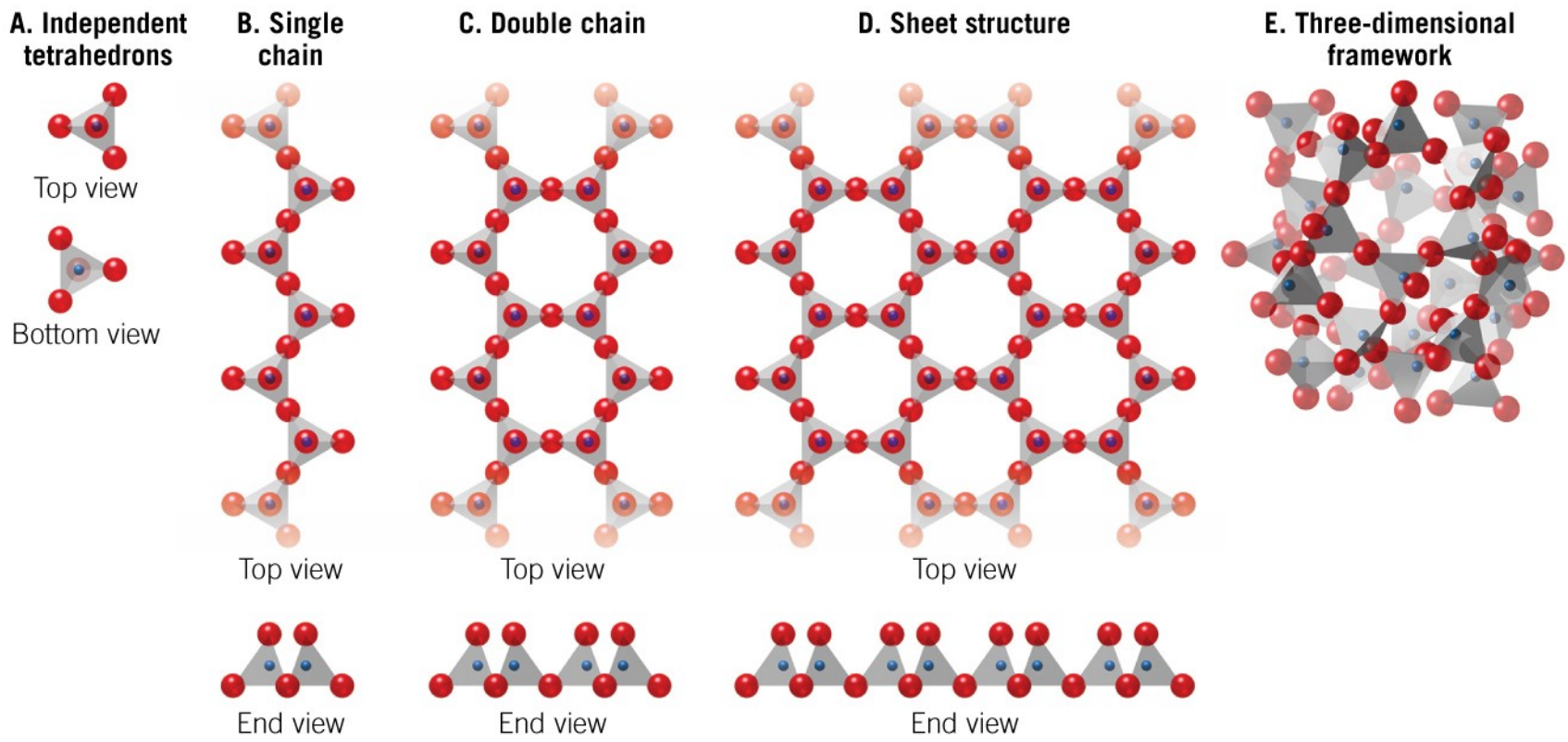

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3.6 The Silicates

- Joining silicate structures
  - Most silicate minerals have a net negative charge (except for quartz)
  - Metal ions are required to balance the charge
  - These positive ions bond with unshared oxygen ions in the tetrahedra
    - Most common ions are Fe^{2+}, Mg^{2+}, K^{+}, Na^{+}, Al^{3+}, Ca^{2+}

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3.6 The Silicates (cont.)

- Joining silicate structures
  - Covalent silicon–oxygen bonds are typically stronger than the ionic bonds of the silicate structure.
  - Controls the cleavage and hardness of minerals
    - Examples:
      - Quartz has a three-dimensional framework, is very hard, and lacks cleavage.
      - Talc has a sheet structure framework bonded with Mg ions and is a very soft mineral.

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3.7 Common Silicate Minerals

- The Light Silicates
  - Generally light in color
  - Specific gravity of approximately 2.7
  - Lack iron and magnesium
  - Contain varying amounts of aluminum, potassium, calcium, and sodium
- The Dark Silicates
  - Generally dark in color
  - Specific gravity between 3.2 and 3.6
  - Contain iron and/or magnesium

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Common Silicate Minerals

Common Silicate Minerals and Mineral Groups

	Mineral/Formula	Cleavage	Silicate Structure	Example
	Olivine group $(Mg,Fe)_2SiO_4$	None	Single tetrahedrons	Olivine
	Pyroxene group (Augite) $(Mg,Fe,Ca,Na)AlSiO_3$	Two planes at 90°	Single chains	Augite
	Amphibole group (Hornblende) $Ca_2(Fe,Mg)_5Si_8O_{22}(OH)_2$	Two planes at 60° and 120°	Double chains	Hornblende
Micas	Biotite $K(Mg,Fe)_3AlSi_3O_{10}(OH)_2$	One plane	Sheets	Biotite
Micas	Muscovite $KAl_2(AlSi_3O_{10})(OH)_2$	One plane	Sheets	Muscovite
Feldspars	Potassium feldspar (Orthoclase) $KAlSi_3O_8$	Two planes at 90°	Three-dimensional networks	Potassium feldspar
Feldspars	Plagioclase $(Ca,Na)AlSi_3O_8$	Two planes at 90°	Three-dimensional networks	
	Quartz SiO_2	None	Three-dimensional networks	Quartz

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3.7 Common Silicate Minerals

- The light silicates
  - Feldspar Group
    - The most common silicate group and makes up more than 50% of Earth's crust
    - Forms under a wide range of temperatures and pressures
    - Exhibits two directions of perfect cleavage at 90 degrees
    - Two most common members:
      - Orthoclase (potassium feldspar)
      - Plagioclase (sodium and calcium feldspar)

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Feldspar Minerals

Potassium Feldspar

A. Potassium feldspar crystal (orthoclase)

B. Potassium feldspar showing cleavage (orthoclase)

Plagioclase Feldspar

C. Sodium-rich plagioclase feldspar (albite)

D. Plagioclase feldspar showing striations (labradorite)

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3.7 Common Silicate Minerals

- The light silicates
  - Quartz
    - The second-most abundant mineral in the continental crust
    - Only common mineral made completely of silicon and oxygen
    - Hard and resistant to weathering
    - Conchoidal fracture
    - Often forms hexagonal crystals
    - Colored by impurities (various ions)

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Quartz

A. Smoky quartz

B. Rose quartz

C. Milky quartz

D. Amethyst

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3.7 Common Silicate Minerals

- The light silicates
  - Muscovite
    - Common member of the mica family
    - Excellent cleavage in one direction
    - Thin sheets are clear
      - Used as glass during the Middle Ages
    - Produces the “glimmering” brilliance often seen in beach sand

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3.7 Common Silicate Minerals (cont.)

- The light silicates
  - Clay minerals
    - “Clay” is a general term used to describe a variety of complex minerals that have sheet structure
    - Clay makes up a large percentage of soil
    - Most originate as products of chemical weathering
    - Kaolinite is common clay mineral used to manufacture fine china

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3.7 Common Silicate Minerals (cont.)

- The dark silicates
  - Olivine group
    - High-temperature silicates
    - Black to green in color
    - Glassy luster and conchoidal fracture
    - Forms small, rounded crystals

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3.7 Common Silicate Minerals (cont.)

- The dark silicates
  - Pyroxene group
    - Important components of dark-colored igneous rocks
    - Augite is the most common mineral in the pyroxene group
      - Black in color
      - Two distinctive cleavages at nearly 90 degrees
      - Dominant mineral in basalt

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3.7 Common Silicate Minerals (cont.)

- The dark silicates
  - Amphibole group
    - Hornblende is the most common mineral in this group
    - Two perfect cleavages exhibiting angles of 120 and 60 degrees

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Augite and Hornblende

A. Augite

B. Hornblende

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3.7 Common Silicate Minerals

- The dark silicates
  - Biotite
    - Iron-rich member of the mica family
    - Excellent cleavage in one direction
  - Garnet
    - Composed of individual tetrahedra linked by metallic ions (similar to olivine)
    - Glassy luster and conchoidal fracture

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3.8 Important Nonsilicate Minerals

- Divided into groups based on the negatively charged ion or complex ion that the members have in common
- Make up approximately 8% of Earth's crust

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3.8 Important Nonsilicate Minerals (cont.)

- Important nonsilicate minerals
  - Several major groups exist including:
    - Oxides
    - Sulfides
    - Sulfates
    - Native elements
    - Carbonates
    - Halides
    - Phosphates

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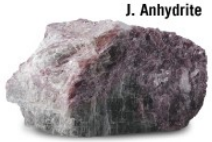
Common Nonsilicate Mineral Groups

Common Nonsilicate Mineral Groups

Mineral Group (key ion(s) or element(s))	Mineral Name	Chemical Formula	Economic Use	Examples
Carbonates (CO_3^{2-})	Calcite Dolomite	$CaCO_3$ $CaMg(CO_3)_2$	Portland cement, lime Portland cement, lime	A. Calcite B. Dolomite
Halides (Cl^{1-}, F^{1-}, Br^{1-})	Halite Fluorite Sylvite	NaCl CaF_2 KCl	Common salt Used in steelmaking Used as fertilizer	C. Halite D. Fluorite
Oxides (O^{2-})	Hematite Magnetite Corundum Ice	Fe_2O_3 Fe_3O_4 Al_2O_3 H_2O	Ore of iron, pigment Ore of iron Gemstone, abrasive Solid form of water	E. Hematite F. Magnetite
Sulfides (S^{2-})	Galena Sphalerite Pyrite Chalcopyrite Cinnabar	PbS ZnS FeS_2 $CuFeS_2$ HgS	Ore of lead Ore of zinc Sulfuric acid production Ore of copper Ore of mercury	G. Galena H. Chalcopyrite
Sulfates (SO_4^{2-})	Gypsum Anhydrite Barite	$CaSO_4 \cdot 2H_2O$ $CaSO_4$ $BaSO_4$	Plaster Plaster Drilling mud	I. Gypsum  J. Anhydrite
Native elements (single elements)	Gold Copper Diamond Graphite Sulfur Silver	Au Cu C C S Ag	Trade, jewelry Electrical conductor Gemstone, abrasive Pencil lead Sulfadrugs, chemicals Jewelry, photography	K. Copper L. Sulfur

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3.9 Minerals: A Nonrenewable Resource

- Renewable: Can be replenished over short time period
- Mineral resources:
  - Already identified deposits that are *currently* recoverable
  - Identified deposits that are not yet economically or technologically recoverable
- Ore
  - A useful metallic mineral that can be mined at a profit
  - Must be concentrated above its average crustal abundance
  - Profitability may change because of economic changes

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Bingham Canyon Copper Mine, Utah

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